In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import sklearn as sk
```

In [24]:
```
df = pd.read_csv(r"C:\Users\Abishek\Downloads\salary_data.csv")
```

In [25]:
```
df.head()
```

Out[25]:

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0

In [26]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     float64
dtypes: float64(2)
memory usage: 612.0 bytes
```

In [27]:
```
df.describe()
```

Out[27]:

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

In [28]:
```
df.shape
```

Out[28]:
```
(30, 2)
```

In [29]:
```
df.duplicated().sum()
```

Out[29]:
```
np.int64(0)
```

In [31]:
```
x=df[['YearsExperience']]
print(x)
```

```
   YearsExperience
0              1.1
1              1.3
2              1.5
3              2.0
4              2.2
5              2.9
6              3.0
7              3.2
8              3.2
9              3.7
10             3.9
11             4.0
12             4.0
13             4.1
14             4.5
15             4.9
16             5.1
17             5.3
18             5.9
19             6.0
20             6.8
21             7.1
22             7.9
23             8.2
24             8.7
25             9.0
26             9.5
27             9.6
28            10.3
29            10.5
```

In [32]:
```
y=df[['Salary']]
print(y)
```

```
     Salary
0   39343.0
1   46205.0
2   37731.0
3   43525.0
4   39891.0
5   56642.0
6   60150.0
7   54445.0
8   64445.0
9   57189.0
10  63218.0
11  55794.0
12  56957.0
13  57081.0
14  61111.0
15  67938.0
16  66029.0
17  83088.0
18  81363.0
19  93940.0
20  91738.0
21  98273.0
22 101302.0
23 113812.0
24 109431.0
25 105582.0
26 116969.0
27 112635.0
28 122391.0
29 121872.0
```

In [36]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=00.2,random_state=12)
```

In [37]:
```
from sklearn.linear_model import LinearRegression
slr=LinearRegression()
slr.fit(x_train,y_train)
slr.predict(x_test)
```

Out[37]:
```
array([[ 55675.5913984 ],
       [100372.57413579],
       [ 81352.58148158],
       [ 68038.58662364],
       [ 37606.59837691],
       [ 92764.5770741 ]])
```

In [38]:
```
from sklearn.metrics import r2_score
y_pred=slr.predict(x_test)
```

```
r2=r2_score(y_test,y_pred)
print(r2)
```

```
0.9429269163491423
```

In [ ]: